



Top 3 Problems

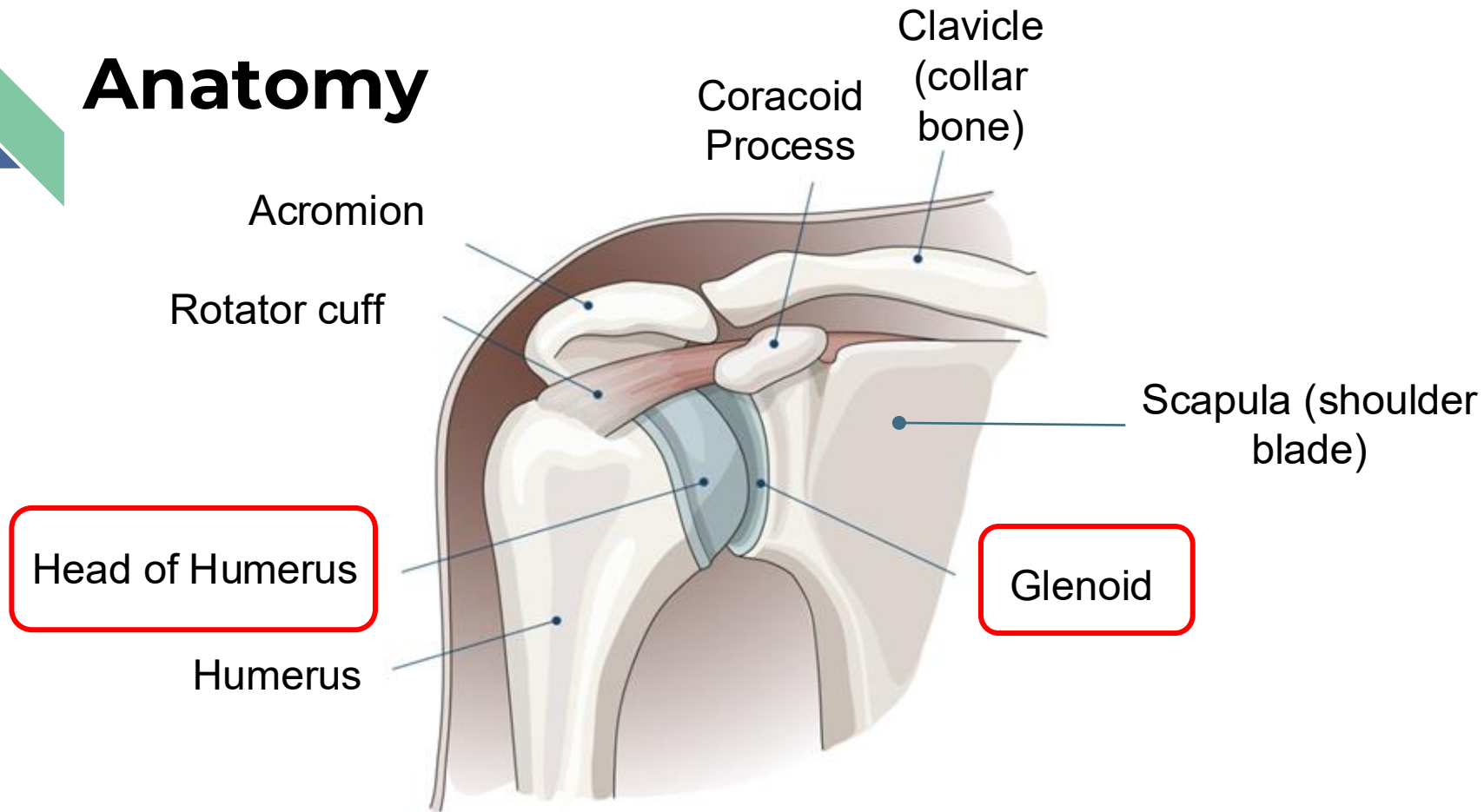
Ortho Team

Bianca, Braxton, Jonathan, Li

Shoulder Replacement Sizing

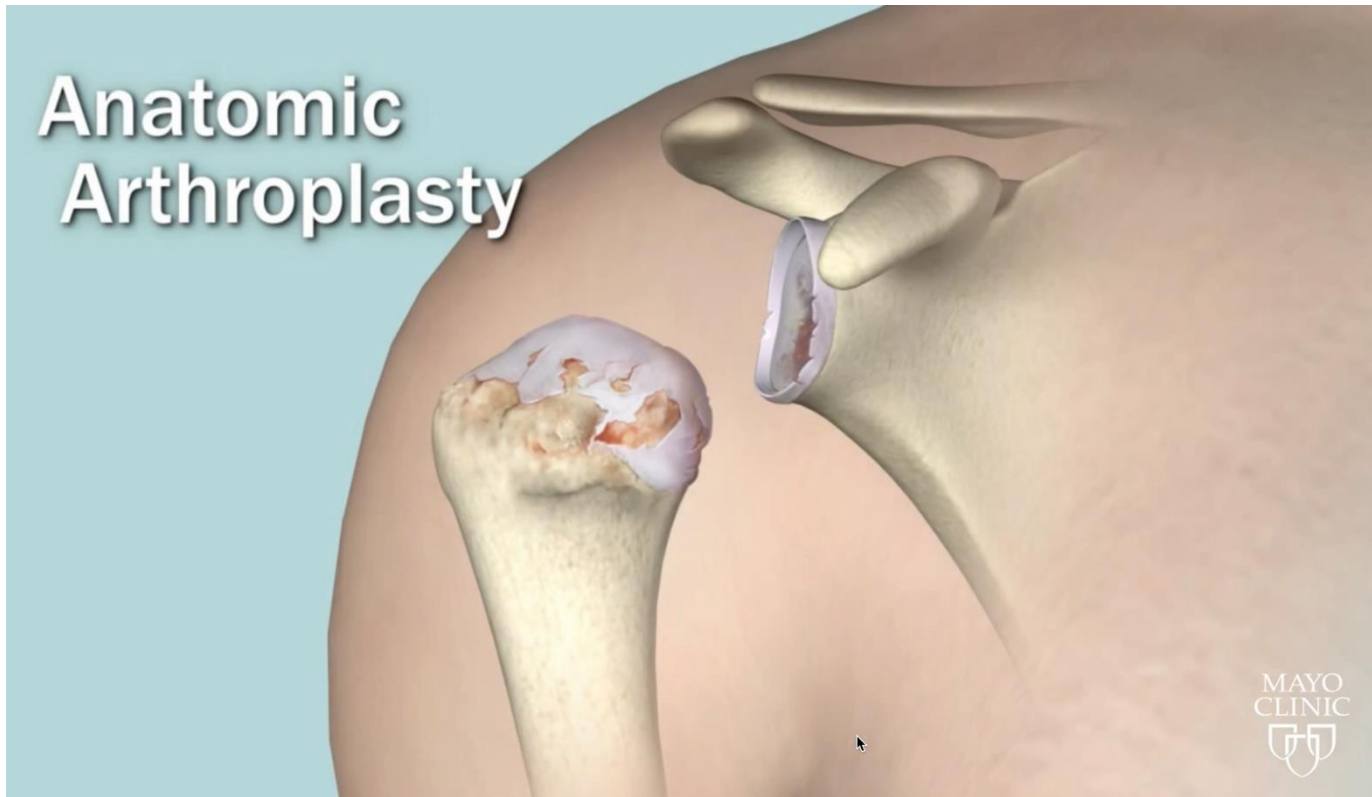


Anatomy



Procedure

Anatomic Arthroplasty

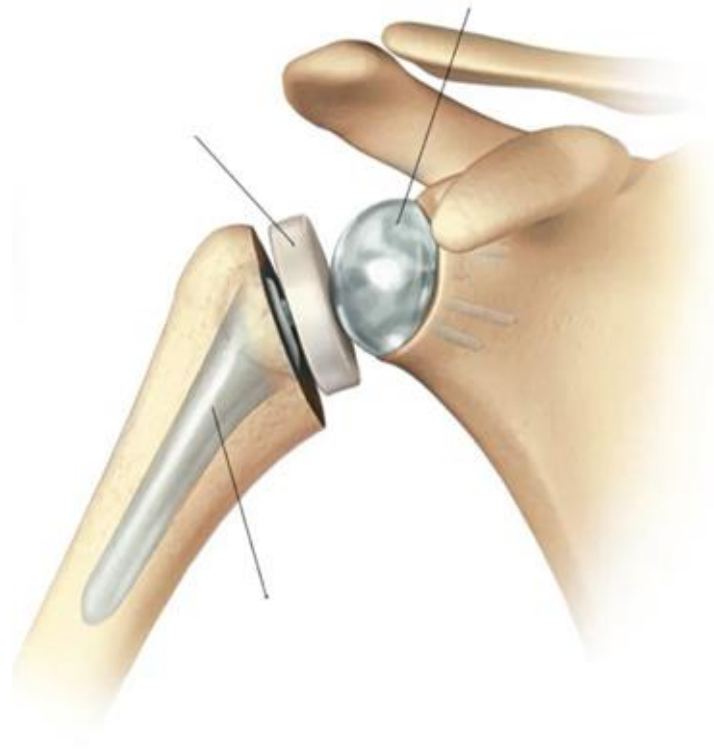


Shoulder Arthroplasty

Anatomical



Reverse





Need Statement

A way to streamline implant size testing for shoulder replacement surgeries, to improve long-term shoulder function and recovery outcomes, while also reducing the need for intraoperative trial-and-error that extends surgery time.

Market Size

Worldwide

- 200,000+

USA

- 50,000

BMC

- 346

Stakeholders:

- Surgeons
- Hospital Administrators
- Patient



Shoulder Arthroplasty Opinions

- Surgical team did not feel like testing takes especially long
- Could take longer to determine optimal size depending on patient anatomy



Shoulder Arthroplasty Key Improvements

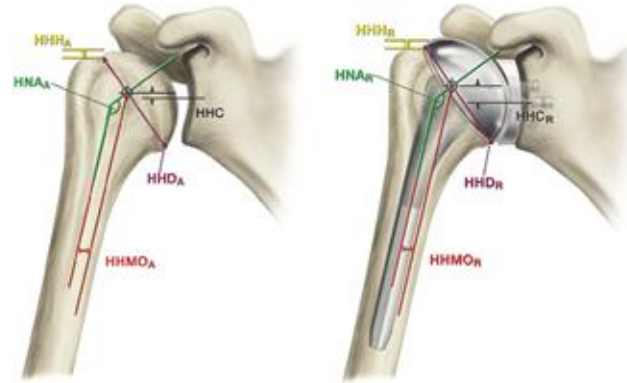
- Implant testing process is slow, especially without extensive pre-op planning
- Several factors make testing process difficult, and it is crucial for surgeons to choose implants that are suitable for the patient



Advantages in streamlining implant sizing in shoulder replacement surgery

Advantages

- Reduced OR time & costs: Saves money
- Patient safety: Less infection risk & anesthesia time
- Implant efficacy: Better fit prevents instability
- Access: Can treat more patients



Products designed to streamline sizing implementation in shoulder arthroplasty

Preoperative Planning Software

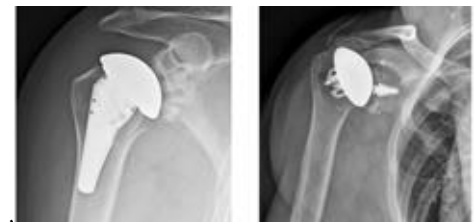
- Concept: CT scans of shoulder → simulate implant fit
- Drawbacks: Requires CT imagery (preoperative planning)
- In practice: Doesn't always reduce OR time (tissue & bone quality impact sizing)

Intraoperative sensor technology

- Concept: Trial implants with sensors → measures forces in glenohumeral joint
- Drawbacks: Learning curve (response to data)
- In practice: Prototype technology (not widely available)

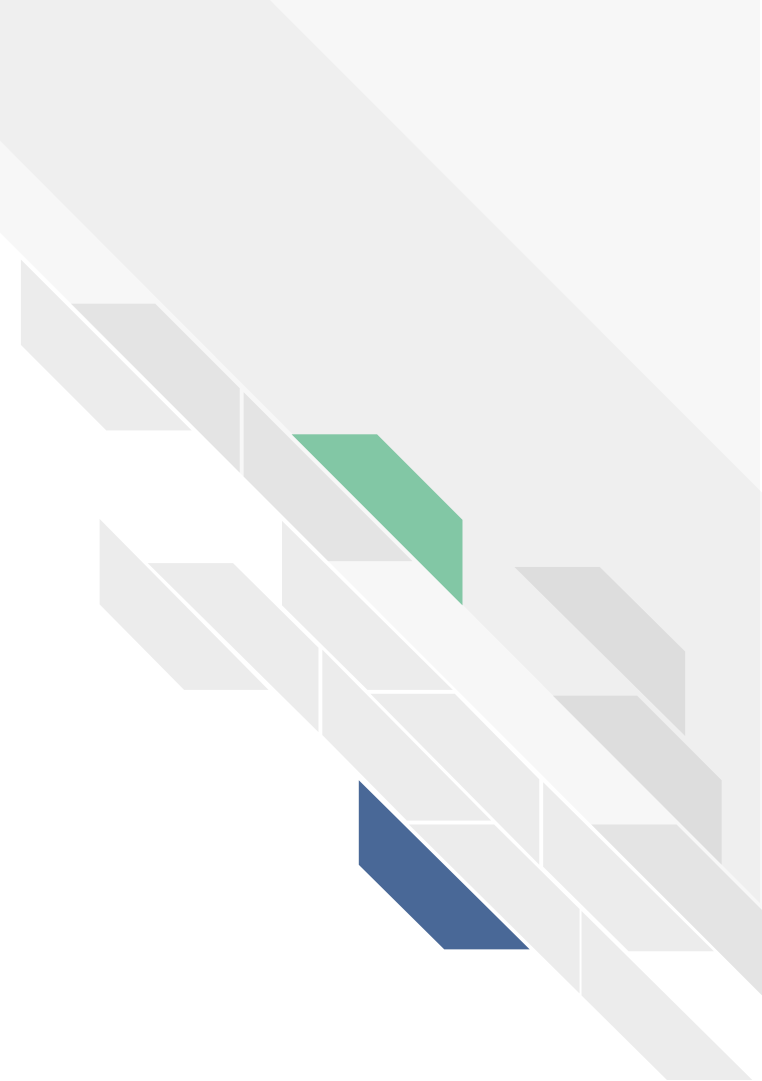
Stemless & Modular Implants

- Concept: Stemless design (removes bone prep) & modular designs (interchangeable parts)
- In practice: Faster trial and error quicker, doesn't eliminate trial phase

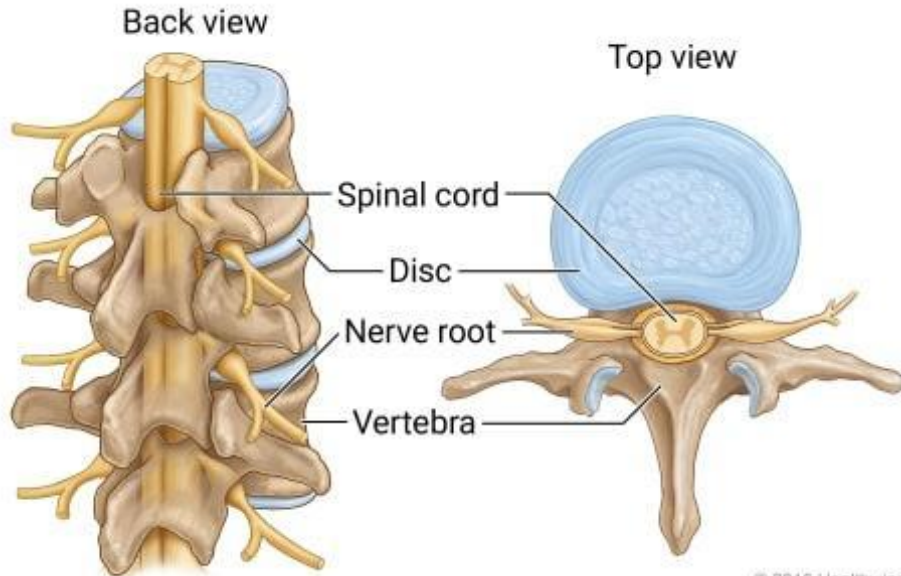


Stemless implant design
(left stem, right stemless)

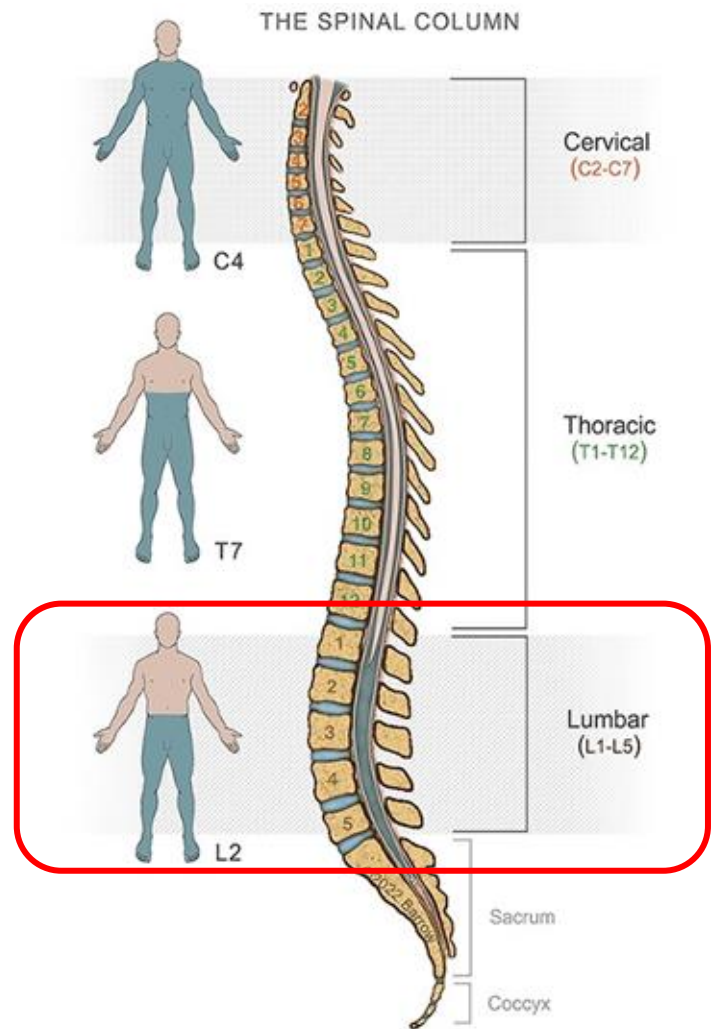
Spinal Fusions - Extreme Lateral Interbody Fusion (XLIF)



Anatomy

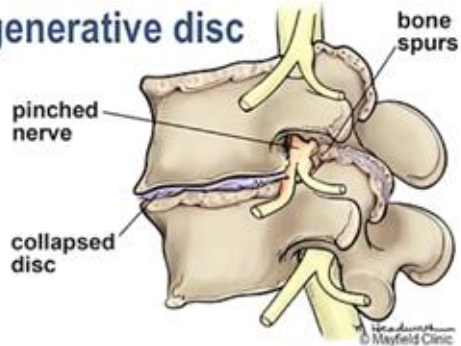


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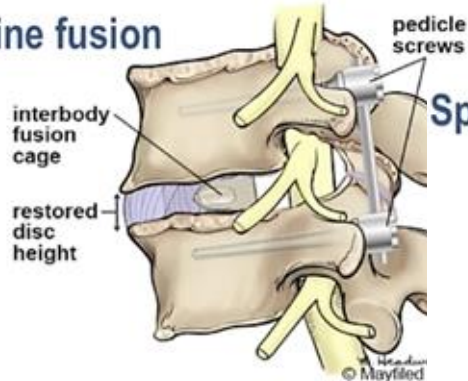


Pathophysiology

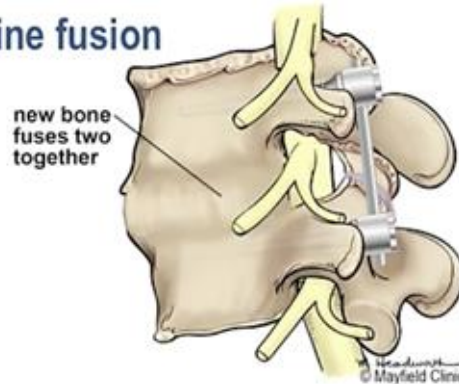
Degenerative disc



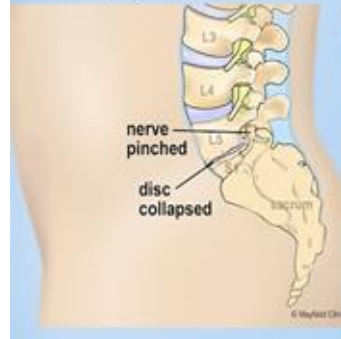
Spine fusion



Spine fusion



Degenerative disc disease



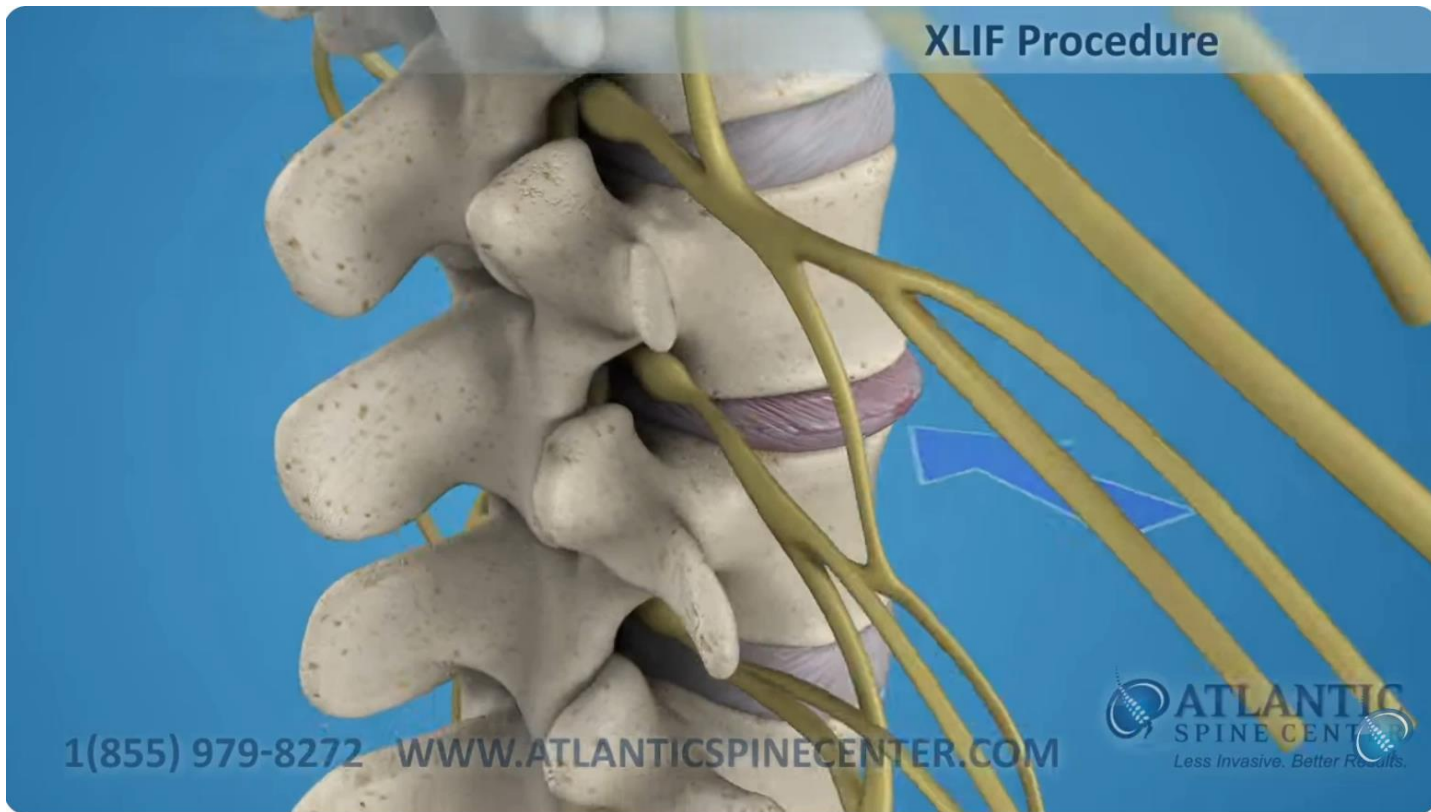
Spondylolisthesis (slip and kink)



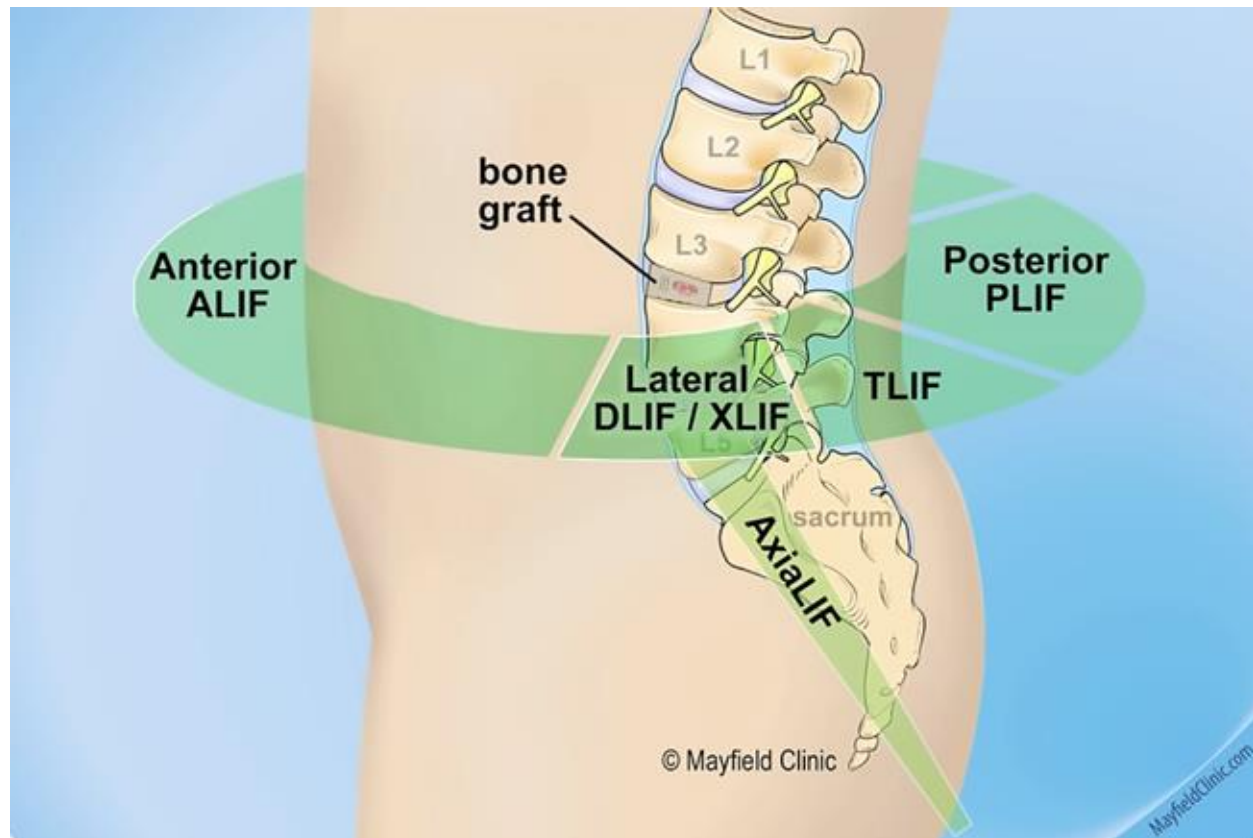
Degenerative scoliosis



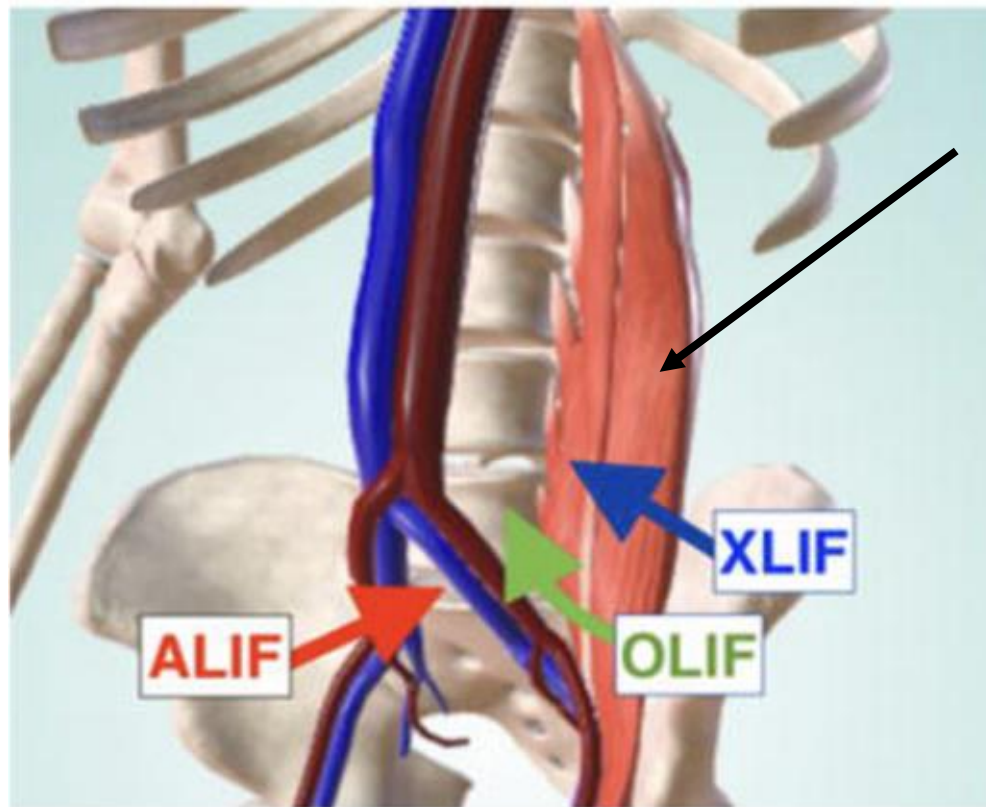
Procedure



Types



Types



Psoas
muscle



Need Statement

A way to prevent nerve root compression during minimally invasive XLIF will reduce rates of nerve complications following procedures

Market Size

Worldwide

- 1.62 million

USA

- 455,500

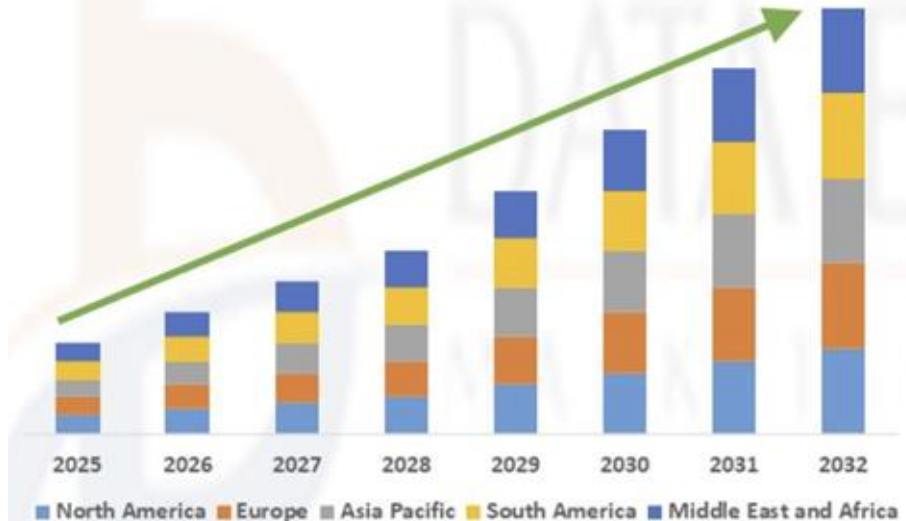
BMC

- 1,000-2,000

Stakeholders:

- Nurses, Patients, Surgeons

Global Extreme Lateral Interbody Fusion (XLIF) Surgery Market is Expected to Account for USD 3.21 Billion by 2032



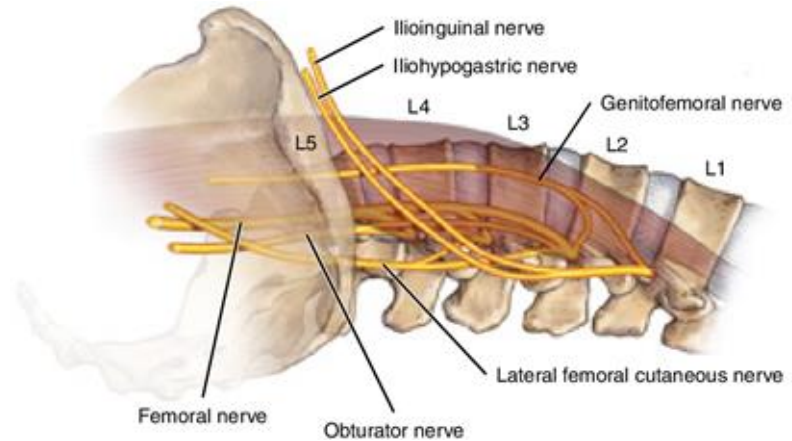
XLIF Opinions

- Steep learning curve for visualization and operation for minimally invasive procedures
- Minimally invasive procedures are generally worth the higher risk of nerve damage if surgeons are experienced enough



XLIF Key Improvements

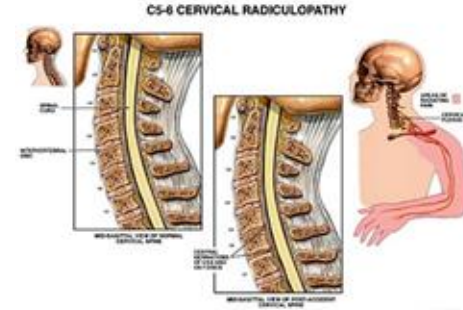
- Make visualization easier so surgeons can more easily navigate around psoas and nerve root
- Find a way to reduce nerve root injury even if surgeons need to operate close to relevant anatomy



Advantages to preventing nerve damage seen in XLIF

Advantages of protecting nerves

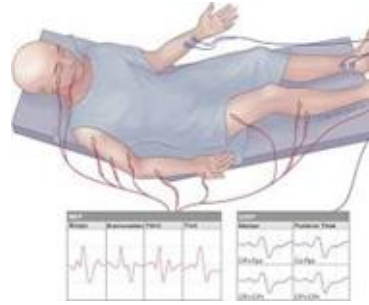
- Patient safety: avoid postoperative nerve pain
- Improved recovery: faster rehabilitation
- Improved surgical efficacy: confidence in successful fusion
- Expanded population: Access to minimally invasive → high nerve risk patients



Existing products designed to prevent nerve damage seen in XLIF

Intraoperative Neuromonitoring

- Concept: EMG-based nerve detection (standard of care)
- Mechanism: Alerts to nerve irritation and compression
- Drawbacks: Lacks alert for prolonged nerve compression



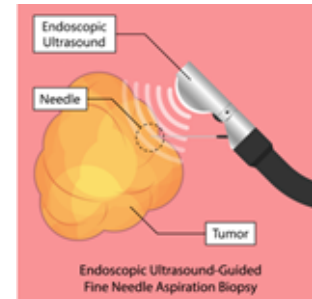
Surgical Retractors

- Concept: Blades to spread the psoas muscle
- Drawbacks: Lead to prolonged muscle retraction and neuropraxia
- Need: Reduce retraction time to promote nerve reperfusion



Endoscopic and ultrasound visualization

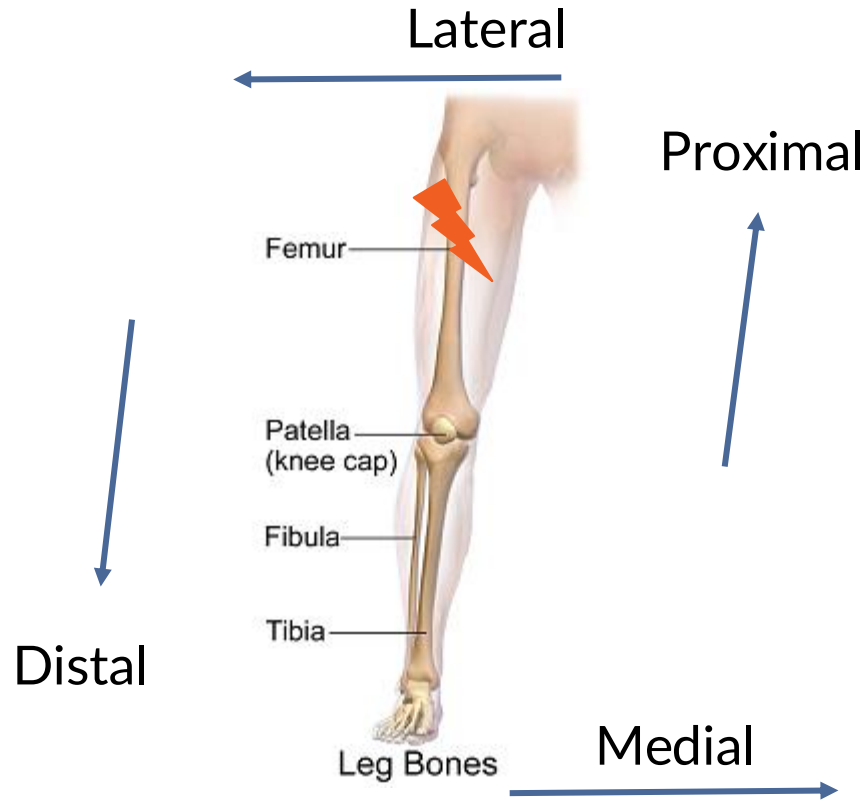
- Concept: Minimally invasive visualization systems (see nerves)
- In practice: Not standard practice in OR



Traction for femoral fractures



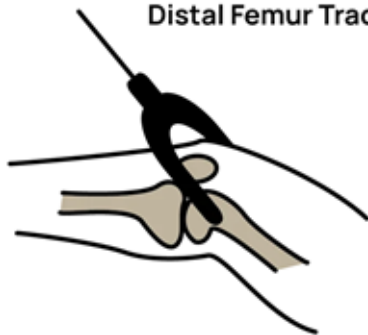
Background



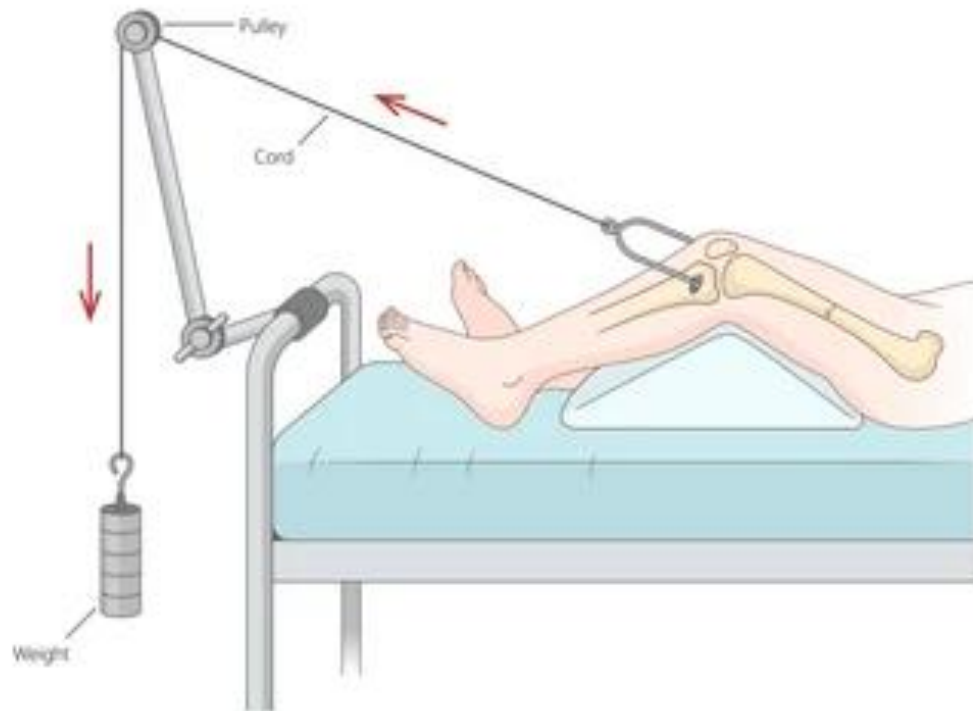
Proximal Tibia Traction



Distal Femur Traction



Skeletal Traction





Need Statement

A way to reduce high-impact trauma femur fractures in young patients that doesn't require direct force on healthy bone.

Market Size

Worldwide

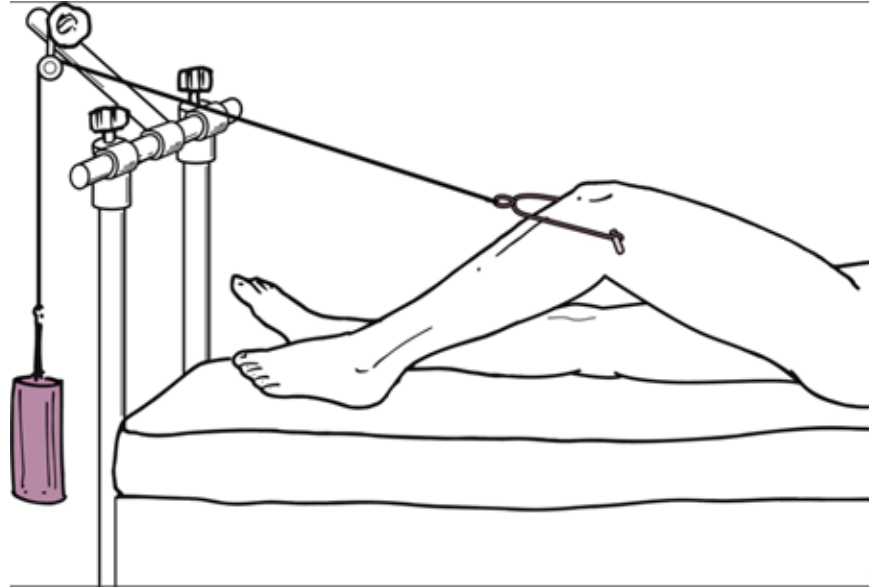
- 1.0 to 2.9 million

USA

- 250,000

Stakeholders:

- Surgeons, Patients, Insurance Companies



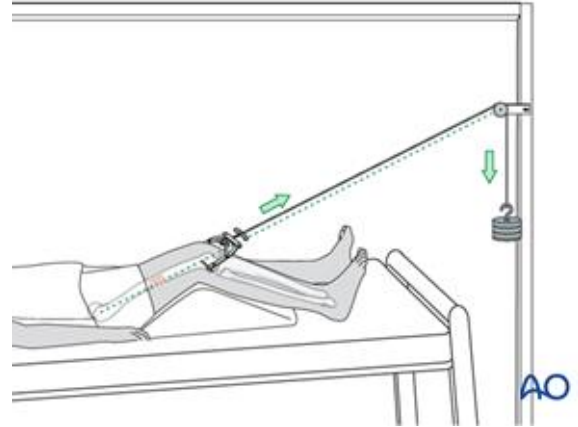
Femur Traction Opinions

- Weights hanging off of the side of the bed are problematic
- Traction alternatives that don't require drilling through healthy bone and direct force



Femur Traction Key Improvements

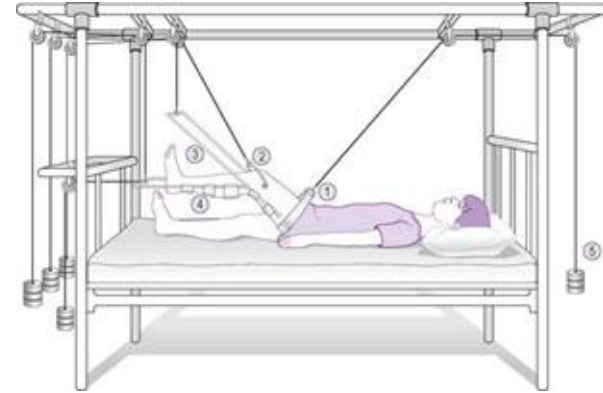
- Weight system could potentially injure surgical team
- Physical and cognitive limitations
- Minimize mechanical force on bone



Advantages of providing improved traction in femoral fracture

Advantages

- Patient safety:
 - Prevents fat embolism
 - Reduces bleeding and pain
- Accelerated treatment: quicker application of traction
- Comfort: Effective design reduces pain during muscle spasm
- Access: User friendly design → easier implementation



Existing products

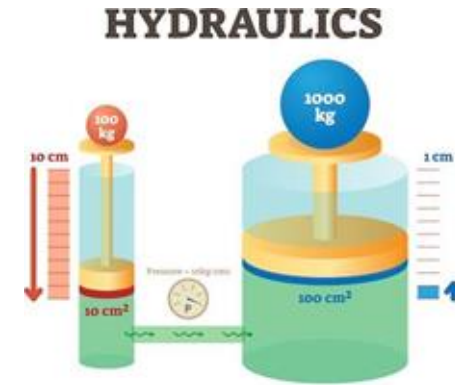


Slishman Traction Splint

- Build: Telescoping pole, traction applied from a strap (no distal bar)
- Concept: Prevents leg overextension & lightweight/compact (fast to deploy)
- Drawbacks: Low tolerance (heavy loads), can't assess tension, expensive

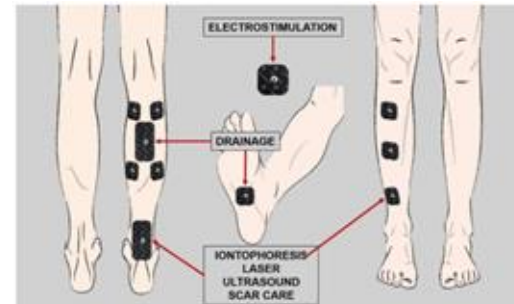
Hydraulic Traction Systems

- Concept: hydraulic cylinders apply traction (maintain tension)
- Advantage: Provides greatly improved precision control
- In practice: Not used due to high cost and complexity



Smart Traction Splints

- Concept: Sensors and microcontrollers track leg tension and adjust tension
- In practice: Not used due to power requirements and cost





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